

ARMessage: An Augmented Reality System for Learning and Practicing Traditional Chinese Medicine Massage

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ABSTRACT

Traditional Chinese Medicine (TCM) massage is well known for its therapeutic effectiveness; however, its training and dissemination are often hindered by a lengthy learning process and high skill-entry requirements. To alleviate these challenges, this paper presents the design and initial implementation of an Augmented Reality (AR)-based system for learning and practicing TCM massage. The proposed system integrates real-time automatic acupoint recognition and calibration, along with visualized feedback of hand-applied pressure on Quest devices, to support more intuitive and guided practice. A preliminary user study was conducted to evaluate the feasibility of the system, through which its current limitations and potential directions for improvement were also identified. The results indicate that incorporating AR technology into TCM massage education can enhance learning efficiency and learner confidence, while providing effective assistance during practical operations. This work demonstrates the potential of immersive technologies to support the modernization and dissemination of TCM massage training.

Index Terms: Augmented Reality, Traditional Chinese Medicine Massage, Acupoint Calibration, Immersive Learning.

1 INTRODUCTION

Modern lifestyles characterized by prolonged sedentary behavior, intensive screen use, and insufficient physical activity have led to a growing prevalence of musculoskeletal discomfort, including neck stiffness and shoulder or back pain. Over time, these conditions may develop into chronic disorders such as frozen shoulder and cervical spondylosis. In this context, Traditional Chinese Medicine (TCM), as a long-established healthcare system, has attracted increasing attention for its potential role in health maintenance and rehabilitation.

As an essential therapeutic modality within TCM, TCM massage has been widely recognized for its effectiveness in the prevention and treatment of chronic conditions, as well as for its low cost and ease of application [1]. However, mastering TCM massage requires systematic training and extensive hands-on practice, resulting in a long learning cycle and a relatively high entry barrier. For novice learners, relying solely on textual materials often proves insufficient. Moreover, the teaching and practice of TCM massage emphasize operational skills, hands-on experience, and individual perception [2, 3]. Consequently, closely integrating learning with practice and introducing emerging technologies may improve learning efficiency and user experience, while also supporting the preservation and dissemination of TCM massage practices.

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Augmented Reality (AR), a representative immersive technology, enables the seamless integration of digital information with the physical world. Kipper and Rampolla [4] defined AR as a technology that combines real-world environments with virtual content. In recent years, AR has been applied across a wide range of domains, including healthcare, education, tourism, design, and manufacturing. In medical and health-related contexts, AR has been explored as a means of embedding guided learning and practical support directly into real-world settings. Although the theoretical and technical frameworks of AR in healthcare remain under development, existing studies suggest its considerable potential for enhancing medical training and practice [5].

At the same time, conventional learning materials often fail to effectively engage learners or sustain attention in health education, which limits the dissemination of relevant knowledge [6]. In the digital era, AR offers an alternative approach by providing interactive, spatially contextualized guidance that supports experiential and practice-oriented learning.

Building on these observations, this study investigates the integration of AR technology with TCM massage learning. Through preliminary user research, we identify key technical factors influencing learning experience and practical effectiveness. An initial prototype system was implemented, and its feasibility and effectiveness were evaluated through early-stage user studies.

2 RELATED WORK

2.1 Innovative Exploration of TCM Massage Teaching and Practice

To preserve and advance TCM massage practices, researchers have explored a range of innovative approaches to teaching and training. With the development of digital technologies, increasing efforts have sought to integrate TCM massage with modern technical tools in order to improve instructional effectiveness and practical outcomes.

In educational contexts, scenario-based and practice-oriented teaching methods have been introduced to enhance learning effectiveness. Shen Mingqiu et al. [7] incorporated simulated patients into massage instruction, integrating theoretical knowledge, technical training, and clinical application to foster practical competence. Ding Yuxin et al. [2] proposed a diagnosis- and treatment-oriented training model that significantly improved student engagement and learning efficiency. While these approaches partially address the limitations of traditional textbook-based learning, they still provide limited intuitive feedback and may involve potential clinical risks during practice.

From the perspective of intelligent systems, existing research has primarily focused on automatic acupoint recognition and calibration, with related applications developed on various intelligent platforms. On mobile devices, camera-based acupoint localization and health applications have been explored. For example, Mei-Ting Su et al. [8] and Lan et al. [9] achieved local acupoint cal-

ibration and designed corresponding healthcare systems. Another line of research employs intelligent robots or robotic arms, using depth cameras to support two-dimensional acupoint calibration and enable precise positioning of back acupoints as well as automated massage operations [10, 11, 12, 13].

Despite these advances in the digitalization and intelligentization of TCM massage, several limitations remain. Most existing teaching approaches emphasize knowledge delivery or acupoint localization, while offering limited immersive or interactive learning experiences. Mobile-based solutions are largely designed for self-care scenarios and are constrained by device performance and usage contexts. In contrast, robotic systems tend to focus on task substitution rather than learner assistance, which restricts opportunities for active skill acquisition.

These limitations highlight the need for new teaching and practice paradigms that reduce learning risks while providing real-time feedback and immersive experiences, thereby enabling a more effective integration of theoretical knowledge and hands-on practice.

2.2 VAMR Integration with Healthcare

Augmented Reality (AR) has demonstrated considerable potential in healthcare education and clinical practice [14]. Medical training based on VAMR technologies has received sustained attention due to its ability to provide immersive, controllable, and repeatable learning experiences [6]. In particular, AR enables the accurate registration of medical imaging data or three-dimensional anatomical models within real environments, supporting intuitive, visual, and interactive learning while reducing risks associated with insufficient practical experience [14, 6].

A growing body of research has demonstrated the effectiveness of AR in medical education. Chen M. developed the AnatoView system, showing that AR-supported visual-spatial cognition facilitates rapid acquisition of anatomical knowledge among non-medical learners [15]. Jose Gutierrez-Maldonado et al. applied VR-based body exposure therapy to improve treatment outcomes for patients with anorexia nervosa [16]. Tao Yu et al. integrated AR displays into surgical microscopes to provide real-time visual assistance, significantly enhancing surgeons' operational performance and confidence [17].

Overall, existing studies indicate that AR-based systems provide experiential learning environments that support the acquisition of practical skills that are difficult to convey through traditional textbooks, classroom instruction, or two-dimensional media. This advantage is particularly relevant to TCM massage, which relies heavily on hands-on practice and experiential learning. Similar to modern medical skills training, TCM massage education faces challenges such as high entry barriers, limited intuitive feedback, and potential safety risks. Therefore, integrating AR into TCM massage teaching can draw on established medical education paradigms to create a safe, immersive, and visually guided training environment, ultimately improving learning efficiency and supporting the sustainable transmission of TCM massage practices.

3 USER RESEARCH

Before the system design phase, we conducted a user needs investigation focusing on learning and practice scenarios that combine TCM massage with AR technology. Participants included five TCM massage learners, three clinically experienced massage practitioners, and six potential learners with an interest in TCM massage. Semi-structured interviews and observational methods were employed to identify key user needs and challenges, which are summarized as follows.

1. **Accuracy of Acupoint Localization:** Most participants reported that accurately locating acupoints is one of the most difficult skills for beginners to master. Although learners may

become familiar with acupoint locations through theoretical study, deviations frequently occur during hands-on practice, and immediate corrective feedback is often unavailable. Practitioners with clinical experience emphasized the importance of *dynamic calibration*, noting that acupoint positions must be adapted to individual differences in body shape and posture.

2. **Perception and Feedback of Applied Pressure:** Learners generally indicated difficulty in judging whether the applied pressure during massage practice is appropriate, often perceiving it as either too light or too forceful. Experienced therapists highlighted pressure control as a key factor distinguishing novices from skilled practitioners. However, current teaching methods provide little objective or quantitative feedback on applied force, relying primarily on verbal instruction from teachers or the learner's subjective perception.
3. **Intelligent and Real-Time Feedback:** Participants expressed a strong desire for real-time guidance during practical operations. Several respondents suggested that the system should provide intuitive prompts when errors occur or assist learners in adjusting their actions through intelligent, context-aware feedback mechanisms.

Overall, the user study indicates that acupoint localization and hand pressure feedback represent the most critical challenges in TCM massage learning and practice. Accurate acupoint calibration addresses beginners' difficulties in locating points correctly, while visualizing applied pressure compensates for the lack of objective feedback in strength training.

4 IMPLEMENTATION

Based on the findings from the preceding analysis, we implemented an initial prototype of an AR-based system for learning and practicing TCM massage. The prototype focuses on two core technical components: real-time calibration of back acupoints and visualization of finger-applied pressure. The AR system was developed using Unity, with the Meta Quest 3 serving as the head-mounted display (HMD). The pressure feedback module was implemented using an Arduino-based hardware platform.

4.1 Real-Time Acupoint Calibration via Passthrough Camera API

The Passthrough Camera API on the Meta Quest 3(S) provides real-time access to the headset's RGB image streams, enabling integration of deep learning-based human keypoint detection in AR. In parallel, the Scene API is used to reconstruct a 3D model of the environment, including the user's body, which serves as the reference for spatial alignment.

During operation, RGB images are sent to a Python module implementing MediaPipe [18] for real-time human keypoint detection. The resulting 2D keypoints in the image plane are then registered with the prebuilt 3D scene using camera parameters and spatial alignment from the Scene API, projecting the detected points into the AR three-dimensional space.

For acupoint localization, the system follows a workflow inspired by prior research on acupoint detection [19]. First, a spatial model is constructed based on the standard set of 17 human pose keypoints. Geometric relationships and mapping rules are then applied to estimate the spatial positions of relevant acupoints, as illustrated in Fig. 1. After obtaining acupoint positions in the image plane, these locations are further projected and mapped onto the reconstructed three-dimensional human model within the AR scene. This completes the real-time calibration and visualization of meridians and acupoints in the AR environment, as shown in Fig. 2.

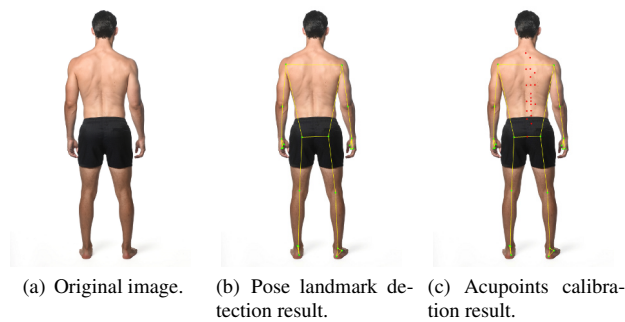


Figure 1: Visualization results of the pose landmark detection and acupoints calibration pipeline.



Figure 2: Calibration result of acupoints on Quest 3S

4.2 Visualization of Finger Pressure Feedback

As shown in Fig. 3, a pressure-sensitive interactive device was developed to capture and visualize hand-applied force during massage practice. The hardware system is based on Arduino/ESP32 microcontrollers and employs Force Sensitive Resistors (FSRs) as the primary sensing components. These sensors are positioned at key force-bearing locations on the palm and fingers to measure pressure applied during interaction.

FSRs are passive components whose resistance varies with applied force and typically exhibit response times below 1 ms, making them suitable for high-frequency sampling. In practice, the achievable sampling rate is constrained by the performance of the microcontroller's analog-to-digital converter (ADC) and the data transmission bandwidth. For example, the Arduino UNO supports ADC sampling rates of approximately 10 kHz, while the ESP32 can reach up to approximately 200 kHz. However, considering serial communication constraints (with stable transmission rates of approximately 200–500 Hz at 115200 baud) and human perceptual requirements for real-time feedback, the sampling frequency in our system was set between 100 and 1000 Hz. On the visualization side, a rendering frame rate of 30–60 fps was sufficient to ensure a smooth interactive experience.

Due to the high sensitivity of thin-film pressure sensors, raw pressure signals are prone to noise and short-term fluctuations. Directly mapping such signals to the visualization interface may result in unstable or jittery feedback. To address this issue, interpolation and smoothing techniques were applied at the data acquisition stage. Specifically, interpolation across consecutive samples combined with moving-average filtering was used to suppress transient noise and reduce signal instability. This processing produced

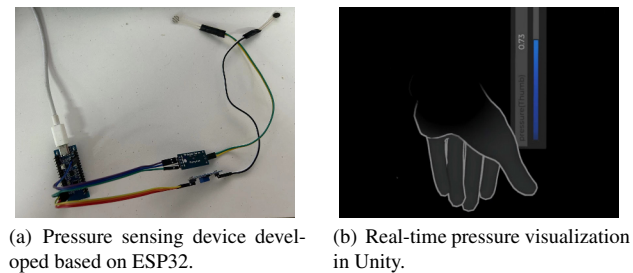


Figure 3: Hardware system and real-time visualization of the pressure sensing platform.

smoother and more continuous pressure curves, thereby improving the clarity and intuitiveness of the visual feedback provided to users.

5 USABILITY EVALUATION

To evaluate the core functionalities of the ARMassage system, we conducted a preliminary usability study to explore user feedback on the interactive experience of integrating AR technology with TCM massage practice.

Two participants (one male and one female) were recruited for the study, assuming the roles of the masseur and the recipient, respectively. Neither participant had received formal training in TCM massage. The participant acting as the masseur had no prior hands-on experience performing TCM massage techniques, while the recipient reported little to no previous experience receiving TCM acupoint massage. In addition, both participants had limited prior exposure to AR systems and head-mounted displays.

Prior to the formal experiment, the research team provided a detailed briefing covering the study objectives, experimental procedure, and basic operational steps of the ARMassage system. With guidance from the experimenter, participants then donned the Meta Quest 3S head-mounted display and entered the system environment.

During the experimental session, participants followed system instructions to complete the initial setup and interaction tasks using the Quest 3S device. The experimenter verbally specified the target acupoints, massage techniques, and intended pressure levels to the participant acting as the masseur, who then performed the corresponding massage operations using the system.

Following the completion of the session, semi-structured interviews lasting approximately 20 minutes were conducted with each participant. Interview responses were recorded in written form. The interview protocol was designed to capture perspectives from both roles and covered multiple dimensions, including the masseur's interactive experience, perceived mental workload during operation, and feedback from the massage recipient regarding comfort and overall experience.

5.1 Result

Feedback from the participant acting as the masseur and wearing the HMD indicated that both the visualized acupoint overlays and the real-time pressure feedback facilitated compliance with the experimenter's instructions during the massage task. The participant reported that these visual cues reduced uncertainty during operation and supported task execution. However, the acupoint recognition results were perceived as less precise than standard acupoint diagrams, and occasional system latency was noted. These issues were reported to increase cognitive load and introduce a degree of psychological pressure during interaction.

Despite these limitations, the participant described the AR-based TCM massage experience as novel and engaging and expressed a willingness to learn and practice TCM massage in an AR-supported environment.

From the perspective of the massage recipient, no pain or discomfort was reported during the session. Instead, the applied pressure was perceived as relatively light. This outcome may be attributed to the pressure feedback mechanism, which enabled the masseur to maintain force within the specified range. At the same time, the masseur's limited prior massage experience and initial unfamiliarity with the system likely constrained fine-grained pressure control.

6 DISCUSSION AND FUTURE WORK

As a preliminary exploration, this study integrates TCM massage training with HMD-based AR technology and proposes a novel interaction paradigm for learning and practice. The current system supports AR-based three-dimensional calibration of back acupoints and visualizes applied pressure through a hand-mounted sensing module, providing learners with intuitive operational guidance and feedback. Preliminary user studies indicate that the system is feasible and has potential practical value for both TCM massage learning and daily practice.

Despite these encouraging results, several technical challenges remain. One notable issue is visual inconsistency in acupoint calibration, as observed in Fig. 1(c) and Fig. 2. These discrepancies primarily result from limitations in the Quest 3S Scene API, which produces relatively coarse reconstructions of human bodies while accurately modeling rigid environmental surfaces. Consequently, projecting 2D-detected acupoints onto the reconstructed 3D scene may introduce spatial offsets. Additional factors such as variations in body type or clothing may further contribute to misalignment. The current implementation also relies on the MediaPipe Pose Landmarker, whose lightweight design enables real-time performance but limits localization accuracy, especially in close-range interactions typical of HMD-based AR.

Currently, the system serves as a proof-of-concept demonstration of AR-based acupoint visualization and interaction. Future work will explore higher-precision 3D reconstruction and depth estimation methods, including dedicated depth sensors (e.g., Kinect) or advanced monocular depth estimation (e.g., Depth Anything [20]). In addition, we plan to adopt parameterized human models such as SMPLX [21] to represent the body, enabling more accurate acupoint visualization and smoother mapping across diverse body types while preserving real-time interaction.

From a user experience perspective, future extensions will incorporate visualized massage technique animations to enhance clarity, gamification elements to improve engagement, and AI-based medical advisor agents to provide personalized guidance, including mappings between user physiological states and massage techniques and customized practice plans. More systematic user studies will be conducted to evaluate learning outcomes, interaction experience, and practical effectiveness, further supporting the integration of TCM massage training with immersive AR technologies.

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